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RESEARCH-ARTICLE

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Dynamic gesture recognition method based on multi-head attention and multi-scale cavity convolution

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Abstract

Since the appearance of depth sensors like Kinect, 3D hand joint coordinate research has expanded. Many studies create gesture recognition systems by calculating joint dependencies, but complex relationships are hard to capture, making features ineffective. This paper proposes an improved model using multi - head attention and multi - scale cavity convolution. The multi - head attention encodes spatial relationships and captures node dependencies. Multi - scale convolution layers capture time information. An attention - weighted loss function boosts the model's robustness. Experiments on the DHG dataset show that for 14 - classified gestures, the model has a 97.8% accuracy, 7% higher than others, and for 28 - classified ones, 92.1% accuracy, 5% higher.

CCS Concepts

• **General and reference** → Cross-computing tools and techniques; Experimentation.

Keywords

Dynamic gesture recognition, spatiotemporal attention, multi-scale dilated convolution, deep learning

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1 Introduction

As human - computer interaction tech advances, gesture recognition is vital for VR, AR, and smart home. Skeleton - based dynamic gesture recognition is a research focus [1], but high - dimensional skeleton data and time - series complexity pose challenges. Traditional methods design hand - crafted features, yet they can't fully capture data in high - dimensional spaces. Different gesture lengths also make feature extraction tough [2]. To address CNNs' limitations, researchers combine them with RNNs (e.g., LSTM). Although CNNs are good at handling noise, combination is still needed. Attention mechanisms, especially self - attention, are beneficial [3]. Some use STA - Res - TCN with attention for recognition, but data issues can affect the model. In short, skeleton - based dynamic gesture recognition has challenges, deep learning offers new paths, and model combination can boost performance [4].

2 Related Work

2.1 Spatiotemporal Positional Encoding

The purpose of positional encoding is to inject signals about relative or absolute positional information into the model's input [13], which is particularly important for processing sequential data. Applying positional encoding to hand joint points allows the network model to better leverage the relationships between joints [14], as shown in Equations 1) to (2).

$$E_{pos,2i} = \sin\left(\frac{pos}{10\quad 000^{\frac{2i}{d}}}\right) \tag{1}$$

$$E_{pos,2i+1} = \cos\left(\frac{pos}{10\quad 000^{\frac{2i}{d}}}\right) \tag{2}$$

In equations, d is encoded feature dimension, pos is gesture's sequence position index, and i is feature dimension index. Sine and cosine provide position info, helping the model learn position relationships. This is crucial in sequence models like Transformers, as it addresses self - attention's lack of position info. Each frame t in the gesture data contains positional information for N joints, represented as $P_t = [p_t^1, p_t^2, \ldots, p_t^N]$, where $p^{\frac{j}{t}}$ is the 3D coordinate vector of the j-th joint in the t-th frame. Assuming that a positional encoding vector E is generated for each joint $p^{\frac{j}{t}}$ at each time step t, it is then combined with the 3D coordinates of the joint to form the enhanced joint feature $F^{\frac{j}{t}}$.

$$\mathbf{F}_t^j = [\mathbf{p}_t^j; \mathbf{E}] \tag{3}$$

In this way, the features of each joint not only contain its positional information in space but also its temporal information in the sequence. The enhanced joint feature representation $F^{\frac{j}{t}}$ is input into subsequent network modules to help the model learn the relative positions and temporal relationships between different joints in the gesture sequence.

2.2 Attention Mechanism

In dynamic gesture recognition, the multi - head attention mechanism, the core of self - attention, significantly boosts the model's feature extraction ability in spatial and temporal dimensions. The design of the multi-head attention mechanism allows the model to process the features of the gesture sequence in parallel across

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multiple subspaces, capturing complex relationships between joints, as shown in Equation 4).

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{Q\mathbf{K}^{\mathrm{T}}}{\sqrt{d_k}}\right)\mathbf{V} \tag{4}$$

In the equation, d_k represents the dimension of the key matrix; Q is the query matrix; K is the key matrix; and V is the value matrix. Dot-product attention typically performs better in terms of speed and space efficiency due to the utilization of highly optimized matrix multiplication [10]. For smaller values of d_k, the performance of both methods is similar; however, when d_k is large, dot-product attention may result in extremely small gradients due to overly large results, which can be addressed by normalization using $\sqrt{d_k}$.

3 Model Architecture

3.1 Spatiotemporal Attention Module

Original node features F from input hand - skeleton coordinates lack spatial ID info (which hand joint a node corresponds to) and temporal info (node's time step). Spatiotemporal positional embeddings, i.e., spatial and temporal ones, are used to add this info. Spatial embeddings have N vectors for hand joints; temporal embeddings have N×T vectors for nodes in the hand - skeleton graph [11]. Node XX's feature vector is combined with corresponding spatial and temporal embedding vectors before subsequent network processing. Thus, we have:

$$\hat{f}_{t,j} = A_{\mathrm{T}}(\mathbf{E}^{\mathrm{T}}_{t,j} + A_{\mathrm{S}}(\mathbf{p}^j_t + \mathbf{E}^{\mathrm{S}}_j)) \tag{5}$$

Where $\hat{f}_{t,j}$ represents the final output characteristics of the node; E^s_j represents the spatial position embedding of the JTH hand joint [5]. $E^T_{t,j}$ means that the JTH hand joint is embedded in the time position of the t time step; A_T and A_S are for handing the location-encoded information to spatial and temporal attention processing operations. These embeddings have the same dimension as the $f_{t,j}$, whose values are set using sine and cosine functions of different frequencies.

The attention section consists of the spatial attention model (AS) and the temporal attention model (AT), based on multi - head attention. They're used to extract spatial and temporal info from the hand skeleton diagram respectively [6]. Finally, the results are aggregated into an average vector, which is used as the feature representation of the skeleton graph for classification. Specifically, given the input feature F^j_t of node p^j_t in the skeleton diagram, the h th spatial attention head first applies three fully connected layers to map F^j_t to key vector, query vector and value vector respectively, as shown in equations 6) ~ (8).

$$\mathbf{K}^h_{t,j} = \mathbf{W}^h_K \mathbf{F}^j_t \tag{6}$$

$$\mathbf{Q}^h_{t,j} = \mathbf{W}^h_Q \mathbf{F}^j_t \tag{7}$$

$$\mathbf{V}^h_{t,j} = \mathbf{W}^h_V \mathbf{F}^j_t \tag{8}$$

Spatial attention computes weights of spatial and self - connection edges in two steps. First, it calculates the scaled dot - product of query and key vectors of nodes in the same time step. Then, it normalizes the results via the softmax function, as in Eqs. (9) - (10).

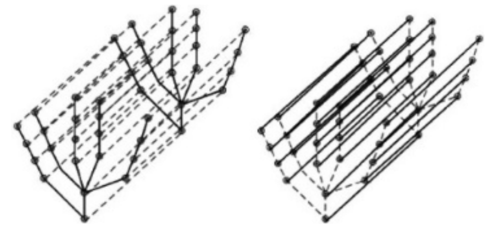

Figure 1: Diagram of masking operation.

$$\mathbf{s}^h_{jk} = \frac{\mathbf{Q}^h_{t,j} \cdot \mathbf{K}^h_{t,k}}{\sqrt{d_{\text{model}}/h}} \tag{9}$$

$$\alpha^h_{jk} = \text{softmax}(\mathbf{s}^h_{jk}) \tag{10}$$

Where: d_{model} is the size of model hidden layer; $\mathbf{s}^h_{jk}$ represents the scaling dot product between node p^j_t and p^k_t; $Q^h_{t,j}$ and $K^h_{t,k}$ are represented by query vector and key vector between skeleton nodes in the same time step respectively [12]. α^h_{jk} is the attention weight between nodes p^j_t and p^k_t, representing the importance of edges between nodes. Then, the spatial attention features of node p^j_t are weighted and averaged with the value vector, and the results are combined, as shown in equations 11) ~ (12).

$$\overline{\mathbf{F}}^h_{t,j} = \sum_{j=1}^{N} (\alpha_{jk} \cdot \mathbf{V}^h_{t,k}) \tag{11}$$

$$\widetilde{\mathbf{F}}^j_t = \text{Concate}[\overline{\mathbf{F}}^1_{t,j}, \overline{\mathbf{F}}^2_{t,j}, \cdots, \overline{\mathbf{F}}^H_{t,j}] \tag{12}$$

Where: H is the number of heads of attention; $\overline{\mathbf{F}}^h_{t,j}$ is the weighted average sum of spatial attention weight of node p^j_t and value vector $\mathbf{V}^h_{t,k}$. $\widetilde{\mathbf{F}}^j_t$ is a representation of spatial attention features that combine the weighted average of multiple heads to form a node p^j_t [7]. For the part of time attention, the output node features of spatial attention model are taken as input, and the attention mechanism mentioned above is adopted to encode. At the same time, the attention is masked in the two dimensions of time and space respectively, so that the network is more focused on the current dimension [8]. The shielding operation diagram is shown in Figure 1.

The specific operation is that given the time step T and the number of joint points J, the spatio-temporal mask matrices are the temporal mask t_{mask} and the spatial mask s_{mask}, with their calculation formulas as follows, respectively:

$$\mathbf{t}_{\text{mask}}[i,j] = \begin{cases} 0 & \text{, iandjareatthesamejointatthesametime} \\ 1 & \text{, othercircumstances} \end{cases} \tag{13}$$

$$\mathbf{s}_{\text{mask}}[i,j] = 1 - \mathbf{t}_{\text{mask}}[i,j] \tag{14}$$

Time masks are for masking irrelevant features at different times, and spatial masks filter in the spatial dimension (between nodes) to mask irrelevant inter - node features [9]. To ensure diagonal

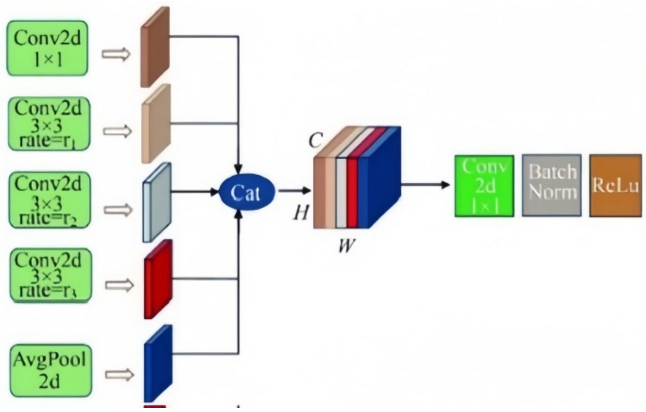

Figure 2: The multi-scale dilated convolution part.

information flow (information transfer of the same node over time), an identity matrix I is added to the mask matrix, i.e.:

$$t_{\text{mask}} = t_{\text{mask}} + I \tag{15}$$

This step ensures that the same node can transmit information in the time dimension. Select the appropriate mask according to the time or space domain for constraint, as shown in equation 16).

$$\alpha = \begin{cases} \alpha \times t_{mask} + (1 - t_{mask}) \times \eta \\ \alpha \times s_{mask} + (1 - s_{mask}) \times \eta \end{cases} \tag{16}$$

Where: η is a minimum value used to exclude irrelevant attention channels so that the activation function output does not consider these positions.

3.2 Fine focusing module

Gestures possess multi - scale temporal characteristics, yet the attention mechanism may yield inaccurate attention distribution due to noise or weight - allocation errors, especially for complex gestures. To tackle this, a fine - focusing module is introduced before the spatial attention mechanism. As shown in Figure 2, it employs multi - scale dilated convolution to adjust dilation rates for extracting features at different granularities. Dilated convolution extends the receptive field to capture multi - scale spatial dependencies with few additional parameters, and is suitable for dealing with inhomogeneity and noise in skeletal joint data.

The multi - scale dilated convolution has five branches. Four convolutional branches with different kernel sizes and dilation rates adapt to various scale features in dynamic gestures. Each branch extracts features from local details to large - scale global info via convolution. The fifth branch extracts global features by global average pooling, ensuring a wider spatial receptive field for the model. Finally, all features are splicing and unified reduction and fusion through convolution to generate the final output, as shown in equation 17).

$$\begin{cases} Y_{1\times1} = \text{ReLU}(\text{BN}(W_{1\times1} * X)) \\ Y_d = \text{ReLU}(\text{BN}(W_{3\times1,d} * X_{dil=d})) \\ Y_G = \text{ReLU}(\text{BN}(W_{1\times1} * \text{GlobalAvgPool}(X))) \\ Y = \text{ReLU}(\text{BN}(W^*_{1\times1}\text{Concat}(Y_{1\times1}, Y_6, Y_{12}, Y_{18}, Y_G))) \end{cases} \tag{17}$$

Where, the void rate d is generated by the input feature, as shown in equation 18).

$$\mathbf{r} = \sigma(\mathbf{W} \cdot \mathbf{P}(\mathbf{X})) \tag{18}$$

P(X) is a pooling op on a feature graph for void - rate context. W is produced by a 1×1 2D convolution mapping features to void rates. By adjusting the void rate, the module can flexibly adapt the receptive field for different inputs, better capturing spatial feature changes.

3.3 Cross entropy loss function based on attention weighting

In this paper, a cross-entropy loss function combined with attention weighting is proposed to improve the classification performance of the model and effectively constrain the attention mechanism.

The cross-entropy loss function is often used in classification tasks and quantifies the difference between the model output probability distribution and the true label probability distribution. For a single sample, y is assumed to be the true distribution, $\hat{y}_i$ is the model output distribution, and the formula for calculating the cross-entropy loss is:

$$loss_{\text{CE}} = -\sum_{i=1}^{n} y_i \ln(\hat{y}_i) \tag{19}$$

Conventional backpropagation uses cross - entropy loss to calculate output - label error and update model parameters. This paper builds on cross - entropy loss by introducing an attention weight penalty mechanism. Its core is to add constraints for misallocated attention, making the model focus on correct classification and enhancing accuracy.

If the maximum attention score $\alpha_{\max}$ is less than 1, the model is not paying enough attention to the correct category. To penalize the model's over - reliance on high attention scores during misclassification, a penalty term P is introduced, expressed as:

$$P = \lambda \sum_{i=1}^{N} \left(1 - \alpha_{\max,i}\right)^2 \tag{20}$$

In the formula, λ is a hyperparameter, which is generally a small value to prevent the penalty item from having too large an impact on the total loss. The final joint loss function comprises cross - entropy loss and a penalty term, i.e.:

$$Loss_{\text{all}} = loss_{CE} + P \tag{21}$$

The design improves classification accuracy with cross - entropy loss and uses an attention weighting mechanism to prevent over - allocation during misclassification, optimizing both classification and attention. It helps the model focus on vital features and boosts performance.

4 Experimental verification

4.1 Experimental data set

DHG, a public dynamic gesture dataset [2], is challenging with 14 right - hand gestures of various finger configs. Captured by Intel Real sense SDK, 20 participants performed each gesture 5 times in 2 configs, collecting 2,800 video seqs (20 - 70 frames). Frames are 3D with 22 - joint skeletons. 9 gestures are coarse - grained, 5

Table 1: The names of the 14 gestures in the DHG dataset.

Serial number	Gesture names and labels	Gesture type
1	Grab (G)	Fine-grained
2	Tap (T)	Coarse-grained
3	Expand (E)	Fine-grained
4	Pinch (P)	Fine-grained
5	Clockwise Rotation (RC)	Fine-grained
6	Counterclockwise Rotation (RCC)	Fine-grained
7	Slide Right (SR)	Coarse-grained
8	Slide Left (SL)	Coarse-grained
9	Slide Up (SU)	Coarse-grained
10	Slide Down (SD)	Coarse-grained
11	Swipe ”x”	Coarse-grained
12	Swipe ”+”	Coarse-grained
13	Swipe ”v”	Coarse-grained
14	Shake (Sh)	Coarse-grained

Table 2: Comparison of recognition rate between the proposed method and other advanced methods in DHG data sets.

Method	Accuracy rate /%	
	14 gesture categories	28 gesture categories
CNN+LSTM	85.60	81.10
ST-GCN	91.20	87.10
MVHAN	92.36	89.56
MBABG	92.00	88.78
MS-ISTGCN	93.70	91.20
Textual method	97.87	93.15

Table 3: Styles available in the Word template.

Heads of attention	Accuracy rate /%	
	14 gesture categories	28 gesture categories
4	96.10	85.71
8	97.87	93.15
10	96.80	89.28
12	96.41	91.42

fine - grained. Dataset’s skeleton info for depth images is used for gesture recognition. Gesture names/types are in Table 1.

4.2 Experimental Setup

Tests were run on NVIDIA GeForce RTX3080. 8 random frames per video were input. The first - frame palm pos. was subtracted from frame sequences, and data was enhanced per Smedt. Trained with Adam optimizer (lr=0.001), batch size 32, dropout 0.2.

4.3 Comparative experiment

This paper experiments on the DHG dataset. To assess the model’s adaptability and effectiveness, it compares the proposed model with mainstream gesture - recognition algorithms. Table 2 shows it outperforms in accuracy, achieving 97.8% for 14 - gesture and 93.1% for 28 - gesture recognition, proving its multi - category superiority and the effectiveness of spatio - temporal attention. The model sets 8 heads in the multi - head attention. Comparative experiments on the head number’s impact are designed, and results are in Table 3.

Experiments indicate that setting the number of attention heads to 8 yields the best model performance and the highest recognition rate. This is particularly true for the 28 - gesture categories, which are highly complex and more sensitive to the number of heads. With eight attention heads, the model can better filter out redundant info and focus on recognition - useful features, reducing misclassification. Having too many or too few heads leads to insufficient or excessive attention to joint relationships, and may cause interference among heads, thus decreasing accuracy.

4.4 Noise interference experiment

To evaluate the network's robustness in practical use, noise with a max amplitude of 0.1 is added to key points in data. A standard convolutional network is used as a reference for comparison under the same input, training, and evaluation. Integrating multi - scale cavity convolution, it enhances multi - scale feature extraction and improves the network's anti - interference in complex scenes.

5 Conclusion

This paper proposes a dynamic gesture recognition network combining multi - scale void convolution and self - attention for skeleton - based tasks. It uses spatial and temporal attention to improve feature extraction, multi - scale cavity convolution for anti - interference, and an attention - based loss function for stability. Experiments prove its effectiveness, but there's room for 28 - gesture recognition improvement. Future research may optimize feature extraction, use multi - scale mechanisms, enhance similar - gesture features, and explore multi - modal fusion.

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